

Daigo Ito

Curriculum Vitae

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Department of Mathematics, Columbia University

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<https://daigo.github.io/>

Research interests Derived categories of coherent sheaves, birational geometry, tensor triangular geometry, homological mirror symmetry

Employment **Columbia University** New York, NY
Joseph F. Ritt Assistant Professor July 2026 - Present

Education **University of California, Berkeley** Berkeley, CA
Ph.D. in Mathematics August 2021 – May 15, 2026
Advisor: David Nadler
Dissertation: Geometry in the Triangular Spectrum of Perfect Derived Categories

Princeton University Princeton, NJ
A.B. in Mathematics September 2017 – May 25, 2021
Advisor: János Kollár, Coadvisor: Joaquín Moraga

Honors and fellowships Ken Ribet & Lisa Goldberg Award in Algebra (Dissertation prize, UC Berkeley) 2026
Simons Foundation Fellowship (Department of Mathematics, UC Berkeley) 2025
Summer Grant Award (Department of Mathematics, UC Berkeley) 2024
Phi Beta Kappa (Princeton University) 2021

Publications 1. **A new proof of the Bondal-Orlov reconstruction using Matsui spectra** (with Hiroki Matsui), *Bull. Lond. Math. Soc.*: Volume 57, Issue 7 (2025), [arXiv version](#).

Preprints 4. **Geometric construction of quiver tensor products** (with John S. Nolan), [arXiv:2510.05277](#) (2025).
3. **A derived category analogue of the Nakai–Moishezon criterion** (with Noah Olander), [arXiv:2507.17681](#) (2025), submitted.
2. **Polarizations on a triangulated category**, [arXiv:2502.15621](#) (2025), submitted.
1. **Gluing of Fourier-Mukai partners in a triangular spectrum and birational geometry**, [arXiv:2309.08147](#) (2023), submitted.

Teaching **Graduate student instructor (University of California, Berkeley)**
Math 52: Calculus Spring 2026

Math 16A: Analytic Geometry and Calculus	Fall 2024
Math 1B: Calculus	Spring 2024
Math 1B: Calculus	Fall 2023
Math 1B: Calculus	Fall 2022
Math 54: Linear Algebra and Differential Equations	Spring 2022
Math 1B: Calculus	Fall 2021

Mentoring

Directed reading program (University of California, Berkeley)

Nathan Han (topics: Introduction to algebraic curves)	Fall 2025
Nathan Mann (topics: Introduction to algebraic geometry and 27 lines)	Spring 2025
Junchang Wang (topics: Riemann surfaces and scheme theory)	Fall 2024
Vivasvat Vatatmaja (topics: Derived category on an elliptic curve)	Spring 2024
Vivasvat Vatatmaja (topics: Model categories and $\mathbb{A}^1$ -homotopy theory)	Spring 2023

Research talks

Conferences & Workshops

- *AMS 2026 Spring Eastern Sectional, Special Session on Derived Categories, Stability Conditions, and Homological Mirror Symmetry*
Boston College, USA March 2026
- *Joint Mathematics Meetings (JMM) 2026, Special Session on Moduli of Surfaces*
Washington, D.C., USA January 2026
- *Western Algebraic Geometry Symposium* (flash talk)
University of Oregon, USA November 2025
- *Derived Representation Theory and Triangulated Categories* (contributed talk)
Aristotle University of Thessaloniki, Greece June 2025
- *Triangulated Categories in Algebra and Geometry* (contributed talk)
Charles University and Czech Academy of Sciences, Czechia September 2024
- *Simple-mindedness, Silting and Stability* (flash talk)
Lancaster University, UK July 2024
- *Classifications of Subcategories of Abelian and Triangulated Categories*
Tokyo University of Agriculture and Technology, Japan June 2024

Seminars

- *D.A.N.C.E. (Derived And Non-Commutative Enthusiasts) Seminar* May 2026
- *Caltech/USC Algebra & Geometry Seminar* April 2026
- *Syzygies and Mirror Symmetry Virtual Seminar* April 2026
- *BIMSA Seminar on Disquisitions on Monoidal Categories and Operads* March 2026
- *UC Santa Barbara Seminar on Geometry and Arithmetic* January 2026
- *UCLA Algebra Seminar* November 2025
- *Stony Brook Algebraic Geometry Seminar* November 2025
- *University of Washington Algebra and Algebraic Geometry Seminar* October 2025
- *UC Berkeley Commutative Algebra and Algebraic Geometry Seminar* October 2025
- *Utah Algebraic Geometry Seminar* September 2025
- *UC Berkeley Commutative Algebra and Algebraic Geometry Seminar* October 2024
- *Tokyo-Nagoya Algebra Seminar* June 2024

	- <i>UC Berkeley Commutative Algebra and Algebraic Geometry Seminar</i>	October 2023
Poster presentations	- <i>Workshop on Mirror Symmetry and Related Topics</i> Kyoto University, Japan	December 2025
	- <i>Western Algebraic Geometry Symposium</i> University of Oregon, USA	November 2025
	- <i>Workshop on Derived Categories, Noncommutative Geometry and Deformation Theory</i> Hotel San Michele, Cetraro, Italy	July 2025
	- <i>Western Algebraic Geometry Symposium</i> The University of Arizona, USA	November 2024
	- <i>Western Algebraic Geometry Symposium</i> University of California, Davis, USA	April 2024
	- <i>SLMath Workshops on Commutative Algebra and Noncommutative Algebraic Geometry</i> SLMath (formerly known as MSRI), USA	Spring 2024
	- <i>Higher Dimensional Algebraic Geometry: James 60</i> University of California, San Diego, USA	January 2024
Organized seminars	<i>Reading group on K3 surfaces and derived categories</i>	Spring 2024
	<i>Geometric Representation Theory Seminar</i> (with Ansuman Bardalai)	Spring 2024
	<i>Geometric Representation Theory Seminar</i> (with Ansuman Bardalai)	Fall 2023
	<i>Reading group on homological mirror symmetry for elliptic curves</i>	Spring 2023
	<i>Reading group on Milnor's characteristic classes</i>	Summer 2022
Skills	Python (SageMath)	